

Supplementary material: Continuous proprioception model

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1 Methods

1.1 General equations for LQG control

The state of the environment $\mathbf{x}_t$ evolves as a function of the previous state and the action $\mathbf{u}_t$ performed by the agent according to a discrete-time linear dynamical system with Gaussian noise

$$\mathbf{x}_{t+1} \sim \mathcal{N}(\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t, \mathbf{\Sigma}_x), \quad (1)$$

whose covariance is $\mathbf{\Sigma}_x$.

At each time step, the agent receives a noisy observation $\mathbf{y}_t$ of the state,

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{H}\mathbf{x}_t, \mathbf{\Sigma}_y), \quad (2)$$

which is a linearly transformed version of the state with additive Gaussian noise with covariance $\mathbf{\Sigma}_y$.

The cost function is quadratic in the state and the actions

$$\sum_{t=1}^T \mathbf{x}_t^\top \mathbf{Q} \mathbf{x}_t + \mathbf{u}_t^\top \mathbf{R} \mathbf{u}_t. \quad (3)$$

This control problem is called linear-quadratic-Gaussian (LQG) control and has a well-known optimal solution, which is composed of optimal state estimation (Kalman filter, KF) and optimal control (linear-quadratic regulator, LQR).

The KF iteratively computes a posterior distribution about the state given a sequence of sensory observations $p(\mathbf{x}_t | \mathbf{y}_1, \dots, \mathbf{y}_t) = \mathcal{N}(\hat{\mathbf{x}}_t, \mathbf{\Sigma}_t)$, where the mean state estimate and its covariance are updated according to

$$\hat{\mathbf{x}}_{t+1} = \mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t + \mathbf{K}_{t+1}(\mathbf{y}_{t+1} - \mathbf{H}(\mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t)) \quad (4)$$

$$\mathbf{\Sigma}_{t+1} = (\mathbf{I} - \mathbf{K}_{t+1}\mathbf{H})\mathbf{P}_{t+1}, \quad (5)$$

where $\mathbf{K}_t$ is defined as

$$\mathbf{K}_{t+1} = \mathbf{P}_{t+1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{t+1} \mathbf{H}^\top + \boldsymbol{\Sigma}_y)^{-1}, \quad (6)$$

$$\mathbf{P}_{t+1} = \mathbf{A} \boldsymbol{\Sigma}_t \mathbf{A} + \boldsymbol{\Sigma}_x. \quad (7)$$

$$(8)$$

The linear-quadratic regulator defines an policy as a function of the current state estimate

$$\mathbf{u}_t = \mathbf{L}_t \hat{\mathbf{x}}_t, \quad (9)$$

which minimizes the expected cost by solving a dynamic programming problem. For a derivation of $\mathbf{L}_t$ and a more detailed derivation of $\mathbf{K}_t$, see e.g. Kochenderfer et al. (2022).

1.2 Target and hand dynamics

We consider two different models of the subject’s hand dynamics, which we call the velocity control model and the point mass model. In both models, the target follows a random walk with variance σ_{rw}^2 and the subject controls the left hand to track the target.

1.2.1 Velocity control model

In the velocity control model, the latent state contains the target position and the subject’s hand position,

$$\mathbf{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \end{bmatrix}, \quad (10)$$

where the target follows a random walk and the subject controls the velocity of the left hand. The generative dynamics are

$$\mathbf{x}_{t+1} = \mathbf{A} \mathbf{x}_t + \mathbf{B} \mathbf{u}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_x), \quad (11)$$

$$\mathbf{A} = \mathbf{I}, \quad \mathbf{B} = dt \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_x = \begin{bmatrix} \sigma_{\text{rw}}^2 & 0 \\ 0 & \sigma_{\text{control}}^2 \end{bmatrix}. \quad (12)$$

The state cost penalizes the tracking error,

$$\mathbf{Q} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{R} = [c]. \quad (13)$$

1.2.2 Point mass model

In the point mass model, the subject’s hand follows a point-mass dynamical system with position, velocity, and force states, while the target still follows a

random walk. In the code, this model is used in the form

$$\mathbf{x}_t = \begin{bmatrix} x_t^{\text{right}} \\ x_t^{\text{left}} \\ v_t^{\text{left}} \\ f_t^{\text{left}} \end{bmatrix}, \quad (14)$$

with the left-hand component obtained from a continuous-time point-mass system (see Kasuga et al., 2022, for details)(see Kasuga et al., 2022). Concretely, the continuous-time dynamics for the hand’s position, velocity and muscle force are

$$m\dot{v}^{\text{left}} = -gv^{\text{left}} + f^{\text{left}}, \quad (15)$$

$$\tau\dot{f}^{\text{left}} = -f^{\text{left}} + u, \quad (16)$$

where g is a damping constant, m the mass, τ the muscle time constant and u the motor command. Writing these equations together with $\dot{x}^{\text{left}} = v^{\text{left}}$ yields a linear continuous-time system with matrices $\mathbf{A}_c$ and $\mathbf{B}_c$. Discretizing this linear system with time step dt (using matrix exponential or standard zero-order hold discretization) yields the discrete-time matrices $\mathbf{A}_{\text{pm}}$ and $\mathbf{B}_{\text{pm}}$. The continuous-time process noise is transformed to the discrete-time covariance Σ_{pm} via Van Loan’s method (Van Loan, 2003).

The full discrete-time dynamics for the whole state are therefore

$$\mathbf{A} = \text{diag}(\mathbf{I}_1, \mathbf{A}_{\text{pm}}), \quad (17)$$

$$\mathbf{B} = \text{diag}(0, \mathbf{B}_{\text{pm}}), \quad (18)$$

$$\Sigma_{\mathbf{x}} = \text{diag}(\sigma_{\text{rw}}^2, \Sigma_{\text{pm}}). \quad (19)$$

The state and action cost matrices for the point mass model penalize tracking error on the positions while leaving velocity and force unpenalized:

$$\mathbf{Q} = \text{diag}\left(\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, 0, 0\right), \quad (20)$$

$$\mathbf{R} = [c]. \quad (21)$$

1.2.3 Delayed sensory observation model

Sensory observations are modelled as the signed difference between the right and left hand positions for each modality $k \in \{\text{prop}, \text{vis}\}$:

$$\mathbf{y}_t^{(k)} = \mathbf{H}\mathbf{x}_{t-\delta_k} + \mathbf{w}_t^{(k)}, \quad \mathbf{w}_t^{(k)} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{y}}^{(k)}), \quad (22)$$

$$\mathbf{H} = \begin{bmatrix} 1 & -1 & 0 & \dots \end{bmatrix}, \quad \Sigma_{\mathbf{y}}^{(k)} = [\sigma_k^2]. \quad (23)$$

To handle sensory delays, we augment the latent state by stacking previous states up to the maximum delay $\delta_{\text{max}} = \max_k \delta_k$ and use the block-companion

form for the dynamics and a block-diagonal state cost (Izawa & Shadmehr, 2008, see):

$$\tilde{\mathbf{x}}_t = [\mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-\delta_{\max}}]^\top, \quad (24)$$

$$\tilde{\mathbf{x}}_{t+1} = \tilde{\mathbf{A}}\tilde{\mathbf{x}}_t + \tilde{\mathbf{B}}\mathbf{u}_t + \tilde{\mathbf{v}}_t, \quad (25)$$

$$\tilde{\mathbf{Q}} = \text{blkdiag}(\mathbf{Q}, \mathbf{0}, \dots, \mathbf{0}). \quad (26)$$

We set $\delta_{\text{vis}} = 2$ and $\delta_{\text{prop}} = 1$, which at $dt = 0.075$ s corresponds to 150 ms and 75 ms sensory delays for vision and proprioception, respectively.

1.3 Model fitting

We fit the model to the behavioral data using inverse optimal control (IOC) for LQG system (Straub & Rothkopf, 2022). More concretely, IOC defines the likelihood of a trajectory of observed states conditional on the model parameters $p(\mathbf{x}_{1:T}^{(i)}|\boldsymbol{\theta})$, where $\mathbf{x}_{1:T}^{(i)}$ is the sequence containing T time steps of observed left and right hand positions in trial i . We assume that the individual trials are independent realizations of tracking behavior generated from the optimal control model with the same parameters $\boldsymbol{\theta}$, s.t. the likelihood of a dataset $\mathcal{D}$ is given by $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_i p(\mathbf{x}_{1:T}^{(i)}|\boldsymbol{\theta})$.

For the experiment with unisensory and multisensory conditions, we assume that the same action cost c and action variability σ_{action} applies across all conditions. We further assume that there is one proprioceptive uncertainty parameter σ_{prop} and two visual uncertainty parameters $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$, one for each condition of visual uncertainty in the experiment.

Overall, the model has five free parameters $\boldsymbol{\theta} = \{c, \sigma_{\text{action}}, \sigma_{\text{prop}}, \sigma_{\text{vis},1}, \sigma_{\text{vis},2}\}$, where $\sigma_{\text{vis},1}$ and $\sigma_{\text{vis},2}$ are the visual noise parameters for the two levels of visual stimuli in the experiment.

We define weakly informative prior distributions over the parameters:

$$\begin{aligned} c &\sim \text{HalfNormal}(1) \\ \sigma_{\text{action}} &\sim \text{HalfNormal}(1) \\ \sigma_{\text{prop}} &\sim \text{HalfNormal}(40) \\ \sigma_{\text{vis},1}, \sigma_{\text{vis},2} &\sim \text{HalfNormal}(40) \end{aligned}$$

To test for optimal integration of visual and proprioceptive information, we consider different versions of the model. They differ only in how the multisensory condition is modeled, while the unisensory (visual and proprioceptive) conditions are always modeled in the same way.

1. In the *proprioceptive dominance* model, only the proprioceptive sensory modality is used in the multisensory condition.
2. In the *equal integration* model, we assume that averaging happens at the level of parameters, such that both sensory modalities use the same param-

eter $\sigma_i = \sqrt{(\sigma_{\text{vis},i}^2 + \sigma_{\text{prop}}^2)/2}$ in the multisensory condition, while unisensory conditions still use their own modality-specific parameters $\sigma_{\text{vis},i}$ and σ_{prop} , respectively.

References

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